2 Fig. 2.1 shows solar-powered traffic warning lights.

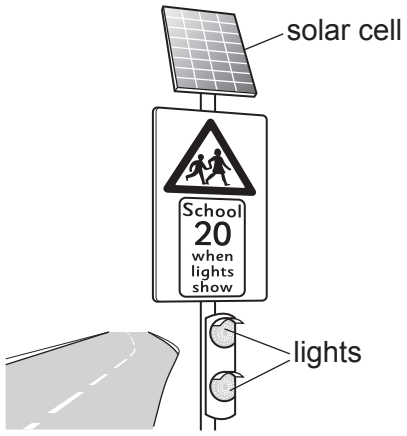


Fig. 2.1

The energy from the solar cell is stored in a battery.

(a) Name the energy store in the battery.

..... [1]

(b) The two lights in Fig. 2.1 are connected in parallel.

State **one** advantage of a parallel connection in a lighting circuit.

.....

..... [1]

(c) The efficiency of the solar cell is 22%. The power supplied to the lights by the cell is 15 W.

(i) State what is meant by 22% efficiency.

.....

..... [1]

(ii) Calculate the solar power input to the solar cell.

power = ... [2]

(d) Suggest **two** advantages of using a solar cell to power the traffic warning lights in Fig. 2.1 compared to using mains electricity.

1 .....

2 .....

[2]

[Total: 7]